A Memory-Guided Agent for Context-Aware Game Localization

COMP7608B Large Language Models Group 50

1 Group members

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2 Project objective

We propose a game-localization agent that translates developer-provided text using imported setting documents, character profiles, scripts, and explicit constraints. The scope is localization from multiple source languages into Simplified Chinese, covering dialogue, UI, quests, skills, items, and system messages. The prototype will support import, editable project memory, translation, developer queries, quality assurance (QA), and export.

We will test whether *selective retrieval from versioned memory and error-driven revision* improves character voice, contextual meaning, terminology, and information preservation under length limits. We will also investigate whether *dependency-based backtracking* can repair earlier translations affected by updated terms or style rules without altering unrelated correct text.

3 Motivation and challenges

Game strings often arrive as isolated fragments, although their meaning depends on speaker, addressee, scene, narrative state, and function. For example, *Ready?* can be a pre-battle question or an interface action. Fluency alone does not establish character-appropriate wording. A *Black Myth: Wukong* study identifies cultural and offensive-language limitations and difficulties evaluating creative translation [5].

Names and messages can recur hundreds of entries later, while homonyms and approved short forms require contextual distinctions. Length limits also demand trade-offs between meaning, terminology, and brevity. Developers will supply these constraints; the text-only system will measure character compliance, not claim verified visual UI fit.

TransAgents motivates project preparation [1], ViDove motivates translation memory [2], and TransGraph motivates selective relational retrieval [3]. We adapt these ideas to game metadata and changing decisions. The agent selects additional retrieval, reuse, compression, developer queries, or backtracking according to evidence and validation results. Developers retain approval control, informed by research on flexible translation assistance [6].

4 Potential solutions

4.1 Architecture and agent roles

The system will use a bounded state machine with an LLM coordinator, specialized workers, and shared storage. The coordinator selects tools and skills from a fixed action set; code validates parameters, enforces budgets, and preserves task state. A heterogeneous model allocation keeps routine processing local and reserves a closed-source API for the translator.

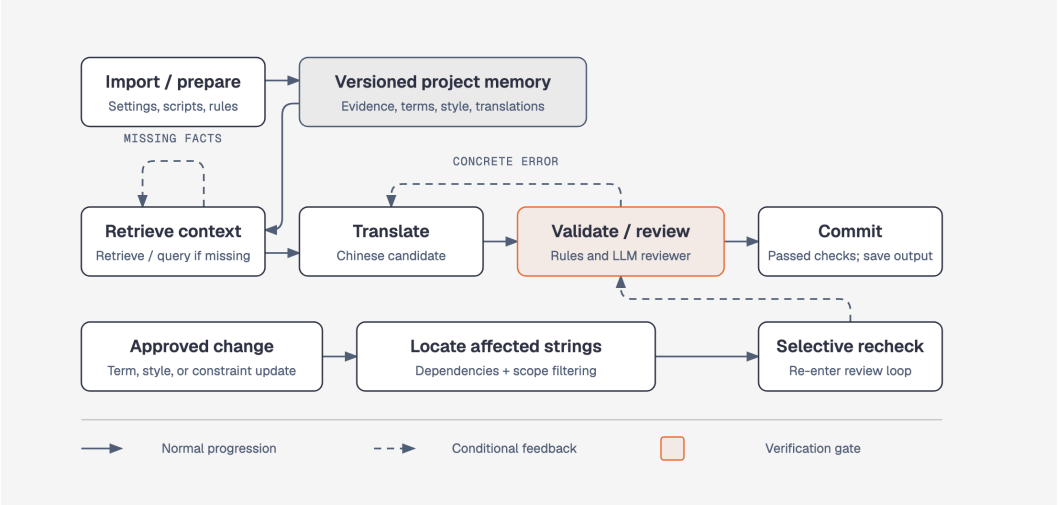


Figure 1: Localization workflow with bounded retrieval, repair, and change-triggered rechecking.

Component	Model allocation	Responsibility and output
Knowledge builder	Local $\leq$7B LLM	Extract entities, relations, terms, and style notes with source evidence.
Context analyst	Local $\leq$7B LLM	Resolve function/state; assemble context packs and identify missing facts.
Translator	Closed-source LLM	Generate/repair Chinese text under project rules and optional length limits.
Reviewer	Local $\sim$27B LLM	Check meaning, voice, terms, and local/global consistency.
Coordinator	Local $\leq$7B LLM	Select tools/skills for retrieval, queries, repair, and backtracking.
Embedding encoder	Small local multilingual model	Encode documents/strings for retrieval.

4.2 Memory, knowledge graph, and retrieval

Source strings carry stable IDs, language labels, and optional type, speaker, addressee, scene, quest, branch, event-condition, and length metadata. Missing context remains unknown. Five memory types—world facts, character/style rules, terminology, translation memory, and scene/QA state—retain provenance, version, scope, and approval status.

Character style notes summarize formality, habitual wording, emotional restraint, and address forms in three to five sentences. Relationship rules are scoped by narrative conditions: later friendship must not determine earlier hostile wording. Source characterization and dialogue ground these notes; generated translations alone cannot establish personality traits.

Term records contain `Source`, `Target`, `Type`, `Context`, `Notes`, and `Status`, plus language, entity/sense IDs, aliases, approved short forms, and versions. Cross-language names share a Chinese term only for a verified common entity. Translation memory stores language pair, source/target, context signature, and review status; reuse requires compatible project, sense, function, relationship state, and constraints.

A lightweight graph links characters, events, scenes, strings, and terms. Relations include SPOKEN_BY, ADDRESSED_TO, REFERS_TO, and PRECEDES. Metadata supplies direct edges; LLM-proposed edges require evidence. Retrieval fetches approved rules/exact terms before graph neighbors and lexical/vector matches to source passages or prior translations. Branch/version filtering and deduplication precede ranking. Translator and reviewer receive the same context pack with evidence IDs and unresolved questions.

4.3 Adaptive tool use and constraint-aware skills

Schema-validated tools support term/style lookup, scene/graph retrieval, translation-memory search, validation, and developer queries. Skills cover character voice, contextual disambiguation, terms, UI compression, consistency audit, and backtrack repair; each defines triggers, inputs, tools, output schema, and acceptance checks. Text-type routing draws on MACD's domain-matched disambiguation [4].

For each scene/batch, the agent retrieves evidence and checks sufficiency. Missing facts that could alter meaning or address forms trigger targeted retrieval, then a developer query specifying the missing field, impact, and provisional alternatives if ambiguity remains. Otherwise, it translates, validates, and reviews; only concrete issues trigger revision. Each string initially permits at most two additional retrieval and two repair rounds. Unresolved cases require developer review.

Developers enable length adaptation per string/group, specifying counting rules, limits, newline policy, and permitted short forms. Unicode grapheme counts, not tokens, measure length. Compression preserves action, negation, conditions, amounts, locked terms, placeholders, and tags; a resurrection button must retain its gold cost. Infeasible requirements trigger a conflict report with proposed abbreviations or relaxed constraints. Dynamic placeholders require developer-provided maximum rendered values; otherwise display length remains unverified.

4.4 Verification and dependency-based backtracking

QA will operate at three scales. Per-string checks cover placeholders, tags, IDs, length, terms, and meaning. Scene-level review examines conversational coherence and character voice. Project-level review groups strings by entity, sense, speaker, and repeated-message signature, then asks the reviewer to distinguish errors from valid contextual variation. It will not place the whole corpus in a single prompt.

Each translation revision will record the versions of the style notes, terms, constraints, and source evidence it used. When a record changes, a reverse dependency index will locate potentially affected translations; an entity/term search will supplement the index to catch omitted dependencies. The coordinator filters these candidates by narrative scope and rechecks them. Changed translations must pass the original validators; failures trigger rollback. Human-approved or locked outputs receive proposed diffs instead of automatic replacement.

For example, approving a short skill name for buttons should recheck related UI entries without replacing the full name in narrative dialogue. A later change to a character's address form should affect only the relevant listener and story phase. This selective backtracking is the primary extension we will test beyond a fixed translate–review pipeline.

4.5 Implementation and economical model deployment

Python tools use validated schemas, SQLite records/graph edges, and a derived vector index. The interface supports CSV/JSON import/export, memory approval, length settings, queries, and revision inspection. Main experiments use Section 4.1's allocation with one shared small local LLM. Quantization, cached embeddings, and selective review reduce costs. A 7B reviewer fallback is a separate option; model choices will be finalized on development data.

The minimum implementation will prioritize style/term memory, adaptive retrieval, rule validation, and backtracking; richer graph extraction is an incremental addition.

5 Experimental dataset description and evaluation plan

5.1 Data construction and change scenarios

We will select open-source game material with settings, contextual scripts, and at least two source languages; *The Battle for Wesnoth* is a candidate [8]. Selection will depend on data and bilingual annotation capacity. We will record revisions and source/license metadata. Settings will form the knowledge corpus; separate annotated strings will drive evaluation. Language versions of the same string will remain in the same split.

We target approximately 600 source strings across the selected languages: 100 development examples and 500 held-out test examples, split by scene where possible. The set will cover three to five characters, several relationship stages, 30–50 terms, and dialogue, UI, quest, skill/item, and system text. Annotators will record speaker, addressee, function, event conditions, acceptable terminology, essential meaning units, and optional length limits. Manually verified synthetic cases may fill gaps and will be reported separately. Counts will be finalized after the data audit; results will be reported per source language as well as overall.

We will construct long-distance recurrence sequences with gaps of 50, 100, and 300 entries, using context-compatible repetitions and contrasting senses. Approximately 20 controlled updates to terminology, style notes, and length constraints will have manually labeled affected/unaffected strings. A separate QA set will contain 60 correct and 60 deliberately corrupted examples covering voice, context, terms, omissions, placeholders, tags, and length. References and QA labels will not be indexed; any seeded approved translation memory will come only from the development set.

5.2 Baselines, ablations, and measurement

We will compare (B0) a human Chinese translation reference, (B1) single-pass translation, (B2) fixed RAG translation, (B3) a fixed retrieve–translate–review pipeline, and (B4) the proposed agent. B1–B4 will share the closed-source translator, source metadata, project rules, and decoding settings. B2–B4 will share the embedding model, imported knowledge, and initial memory; B3–B4 will use the same local reviewer. B4 adds local planning and adaptive actions. We will report tool/model calls and matched-budget results on a subset. Update tests will compare targeted repair with full retranslation.

Verified human-authored Chinese localizations will provide B0; where unavailable, bilingual translators will prepare references for the assessed subset under the project brief. They will be scored with the same rubric as system outputs and excluded from retrieval memory. Alternative valid translations remain acceptable.

Dimension	Planned measurements
Translation quality	Blinded bilingual ratings of accuracy, contextual meaning, character voice, and naturalness; pairwise preference.
Consistency	Entity/sense-specific term accuracy and consistency across context-compatible repetitions, stratified by recurrence gap.
QA and backtracking	Error detection precision/recall/F1; affected-string retrieval precision/recall; correction and unnecessary-edit rates.
Practical cost	Tokens and tool calls, latency, API charges, and local GPU time/peak memory.

Two bilingual raters qualified for the relevant source language will independently assess a stratified subset of approximately 120 test strings under a shared rubric, with agreement and adjudicated disagreements reported. Reference authors will not rate their own translations. Automatic reference-based scores will be supplementary because multiple creative translations can be valid [5]. Following game-LQA benchmarking, detection quality will be evaluated on both clean and faulty cases rather than recall alone [7].

Ablations will remove character style notes, graph-assisted retrieval, or dependency-based backtracking, while retaining the remaining workflow to examine each component's contribution.

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